

Common Infectious Diseases Reference

Prepared for Health Assist AI RAG knowledge base

This document is a concise, source-grounded reference compiled from official Government of India and WHO material. It is designed for retrieval and educational answers, not for diagnosis or prescribing. Drug doses are intentionally omitted.

Coverage

Chickenpox, measles, dengue, malaria, chikungunya, typhoid, cholera/acute watery diarrhoea, viral hepatitis, tuberculosis, leptospirosis, rabies, diphtheria, pertussis, influenza-like illness, and acute respiratory infection/pneumonia.

1. Chickenpox (Varicella)

Cause

Varicella-zoster virus (VZV).

How it spreads

Highly contagious. It spreads through respiratory droplets/aerosols and direct contact with fluid from skin blisters. A person can be contagious before the rash appears and until lesions have crusted.

Common symptoms

Fever, malaise and reduced appetite may occur before an itchy rash. The rash typically progresses from flat spots to raised bumps, fluid-filled blisters and then scabs, often appearing in crops on the face/scalp and trunk before spreading.

Warning signs / when to seek urgent care

Seek medical care promptly for breathing difficulty, severe weakness, confusion, dehydration, a rapidly worsening rash, signs of bacterial skin infection, or illness in pregnancy, newborns, adults, or people with weakened immunity.

Prevention

Vaccination where recommended, avoiding close contact with infected people, respiratory hygiene, and keeping an infected person away from susceptible people until lesions have crusted.

General management

Most uncomplicated cases are managed supportively with fluids, rest, itch relief and fever control. Medical assessment is important for people at higher risk of complications.

Primary sources: WHO Chickenpox Q&A (2025); MoHFW Standard Treatment Guidelines.

2. Measles

Cause

Measles virus.

How it spreads

An extremely contagious airborne infection spread when an infected person breathes, coughs or sneezes.

Common symptoms

High fever, cough, runny nose, red/watery eyes, small white spots inside the cheeks, followed by a widespread rash.

Warning signs / when to seek urgent care

Urgent assessment is needed for breathing difficulty, severe dehydration, persistent drowsiness/confusion, seizures, or serious illness in a young child or immunocompromised person.

Prevention

Vaccination is the main preventive measure. Isolation of suspected cases and good respiratory hygiene reduce transmission.

General management

Treatment is mainly supportive and complications should be assessed by a clinician. Suspected measles should be medically evaluated, especially in children.

3. Dengue

Cause

Dengue virus, transmitted mainly by infected Aedes mosquitoes.

How it spreads

Primarily through the bite of infected Aedes mosquitoes, which commonly bite during the day.

Common symptoms

High fever, severe headache, pain behind the eyes, muscle/joint aches, nausea, vomiting, swollen glands, fatigue and rash.

Warning signs / when to seek urgent care

Severe abdominal pain, persistent vomiting, rapid breathing, bleeding from nose/gums, blood in vomit or stool, marked weakness, restlessness, very thirsty feeling, or pale/cold skin require immediate medical care.

Prevention

Prevent mosquito bites and breeding: use repellents, protective clothing/screens, and regularly empty or cover water-holding containers.

General management

Rest and fluids are important. WHO advises paracetamol/acetaminophen for fever or pain and avoiding aspirin/ibuprofen when dengue is suspected because of bleeding risk. Severe symptoms need urgent care.

Primary sources: WHO Dengue fact sheet (2025); MoHFW National Guidelines for Clinical Management of Dengue Fever.

4. Malaria

Cause

Plasmodium parasites transmitted by infected Anopheles mosquitoes.

How it spreads

Through bites of infected female Anopheles mosquitoes.

Common symptoms

Fever, headache and chills are common early symptoms. Symptoms can initially be mild but falciparum malaria can progress rapidly.

Warning signs / when to seek urgent care

Confusion, seizures, breathing difficulty, severe weakness, jaundice, repeated vomiting, inability to drink, or deterioration in a pregnant person or young child requires urgent care.

Prevention

Mosquito nets, repellents, protective clothing, vector control and appropriate preventive medicines for eligible travellers or high-risk groups when prescribed.

General management

Malaria requires diagnostic testing and prompt antimalarial treatment selected by a qualified clinician. Do not rely on home treatment alone when malaria is suspected.

Primary sources: WHO Malaria health topic; MoHFW/NVBDCP Diagnosis and Treatment of Malaria.

5. Chikungunya

Cause

Chikungunya virus (CHIKV).

How it spreads

Transmitted mainly by infected *Aedes aegypti* and *Aedes albopictus* mosquitoes.

Common symptoms

Abrupt fever with severe joint pain; joint swelling, muscle pain, headache, nausea, fatigue and rash can occur. Joint pain can persist for weeks or longer in some people.

Warning signs / when to seek urgent care

Medical assessment is especially important for newborns, older adults, people with underlying disease, or anyone with severe weakness, neurological symptoms, chest symptoms or inability to maintain hydration.

Prevention

Avoid mosquito bites and reduce standing-water breeding sites. A person with suspected chikungunya should also avoid mosquito bites during the first week of illness.

General management

Supportive care includes rest, fluids and fever/pain management. Until dengue is excluded, WHO recommends paracetamol/acetaminophen rather than NSAIDs because of possible bleeding risk.

Primary sources: WHO Chikungunya fact sheet (2025); MoHFW/NVBDCP Chikungunya Fever guidance.

6. Typhoid (Enteric Fever)

Cause

Salmonella Typhi bacterium.

How it spreads

Usually through food or water contaminated by faeces from an infected person or carrier.

Common symptoms

Prolonged fever, fatigue, headache, nausea, abdominal pain, poor appetite, constipation or diarrhoea; some people develop a rash.

Warning signs / when to seek urgent care

Severe abdominal pain, gastrointestinal bleeding, confusion, persistent vomiting, marked weakness or signs of shock require urgent hospital assessment.

Prevention

Safe drinking water, sanitation, handwashing, safe food preparation and typhoid vaccination where recommended.

General management

Diagnosis commonly involves clinical assessment and laboratory testing. Typhoid is treated with antibiotics chosen by a clinician because antimicrobial resistance is an important concern.

Primary sources: WHO Typhoid fact sheet; MoHFW National Treatment Guidelines for Antimicrobial Use in Infectious Diseases.

7. Cholera and Acute Watery Diarrhoea

Cause

Cholera is caused by *Vibrio cholerae*. Acute diarrhoea can have many infectious causes.

How it spreads

Cholera spreads through food or water contaminated with the bacterium, especially where safe water and sanitation are inadequate.

Common symptoms

Sudden watery diarrhoea is typical of cholera. Vomiting and rapid fluid loss may occur; many infections are mild but severe disease can cause dangerous dehydration within hours.

Warning signs / when to seek urgent care

Very frequent watery stools, inability to drink, lethargy, sunken eyes, very little urine, weak pulse, cold extremities or altered consciousness require urgent treatment.

Prevention

Safe water, sanitation, handwashing, food hygiene and, in appropriate settings, oral cholera vaccination.

General management

Rapid rehydration is the cornerstone. Oral rehydration solution is used for many cases; severe dehydration requires urgent medical care and may need intravenous fluids and clinician-directed antibiotics.

Primary sources: WHO Cholera fact sheet (2024); Indian public-health communicable disease guidance.

8. Viral Hepatitis

Cause

Several viruses can cause hepatitis, including hepatitis A, B, C, D and E viruses.

How it spreads

Route depends on the virus. Hepatitis A and E are commonly linked to contaminated food/water; B and C can spread through infected blood and certain body fluids.

Common symptoms

May include fatigue, nausea, abdominal discomfort, dark urine, pale stools and jaundice, although some infections have few or no symptoms.

Warning signs / when to seek urgent care

Increasing drowsiness/confusion, repeated vomiting, bleeding, severe abdominal symptoms or rapidly worsening jaundice need urgent assessment.

Prevention

Safe water and food, safe injections and blood practices, avoiding shared needles, safer sex, and vaccination for hepatitis A/B where indicated.

General management

Testing is needed to identify the type of hepatitis. Management differs substantially by virus, so treatment should be clinician-directed.

Primary sources: MoHFW National Guidelines for Diagnosis & Management of Viral Hepatitis; NCDC communicable-disease resources.

9. Tuberculosis (TB)

Cause

Mycobacterium tuberculosis bacteria.

How it spreads

Airborne spread, particularly during prolonged close contact with a person who has infectious pulmonary TB.

Common symptoms

Persistent cough, fever, night sweats, weight loss, fatigue and sometimes coughing up blood.

Warning signs / when to seek urgent care

Coughing up significant blood, severe breathlessness, confusion, or severe deterioration requires urgent care.

Prevention

Early diagnosis and complete treatment reduce spread. Ventilation, cough hygiene, contact evaluation and national TB-control measures are important.

General management

TB requires proper diagnostic testing and a full multi-drug treatment regimen under medical supervision. Treatment should not be stopped early.

Primary sources: MoHFW Standard Treatment Guidelines / National TB programme resources; NCDC.

10. Leptospirosis

Cause

Leptospira bacteria.

How it spreads

Exposure to water, soil or food contaminated with urine from infected animals, often through cuts/abrasions or mucous membranes.

Common symptoms

Acute fever, headache, muscle pain, chills, red eyes, nausea/vomiting and sometimes jaundice. Symptoms can overlap with dengue, malaria and other febrile illnesses.

Warning signs / when to seek urgent care

Jaundice, reduced urine, bleeding, breathing difficulty, confusion or severe weakness can indicate serious disease and require urgent care.

Prevention

Avoid potentially contaminated floodwater when possible, use protective footwear/gloves, cover wounds, and improve rodent/environmental control.

General management

Suspected leptospirosis needs medical evaluation and may require antibiotics and supportive treatment, especially when severe disease is possible.

Primary sources: MoHFW National Guidelines for Diagnosis, Case Management, Prevention and Control of Leptospirosis.

11. Rabies

Cause

Rabies virus.

How it spreads

Usually through bites or scratches from an infected mammal when saliva enters broken skin or mucous membranes.

Common symptoms

Early symptoms may include fever and pain or unusual sensations at the wound site. Once neurological symptoms develop, rabies is almost always fatal.

Warning signs / when to seek urgent care

Any potentially rabid animal bite or scratch is an urgent medical issue. Do not wait for symptoms.

Prevention

Immediate wound washing and prompt post-exposure prophylaxis when indicated. Vaccinating dogs is central to preventing human rabies.

General management

Wash the wound thoroughly with soap and running water and seek medical care urgently for assessment of rabies vaccine and, when indicated, rabies immunoglobulin.

Primary sources: MoHFW National Guidelines on Rabies Prophylaxis; NCDC National Rabies Control Programme.

12. Diphtheria

Cause

Toxin-producing *Corynebacterium diphtheriae* bacteria.

How it spreads

Mainly respiratory droplets and close contact; sometimes contact with infected skin lesions.

Common symptoms

Sore throat, fever, swollen neck glands and a thick grey/white membrane in the throat can occur.

Warning signs / when to seek urgent care

Breathing or swallowing difficulty, neck swelling, weakness or suspected throat membrane requires urgent hospital assessment.

Prevention

Routine vaccination is the main preventive measure. Suspected cases require public-health and contact-management measures.

General management

Diphtheria is a medical emergency requiring clinician-directed antitoxin and antibiotics; treatment should not be delayed when strongly suspected.

Primary sources: NCDC communicable-disease resources; MoHFW vaccine-preventable disease surveillance guidance.

13. Pertussis (Whooping Cough)

Cause

Bordetella pertussis bacterium.

How it spreads

Respiratory droplets during close contact.

Common symptoms

Often begins like a common cold, then progresses to repeated coughing fits. A 'whoop', vomiting after coughing or pauses in breathing may occur, especially in children.

Warning signs / when to seek urgent care

Breathing pauses, blue coloration, severe breathing difficulty, dehydration, seizures, or illness in a young infant require urgent care.

Prevention

Routine vaccination and appropriate management of close contacts reduce transmission.

General management

Medical assessment is important, particularly for infants and pregnant contacts. Antibiotics are used in appropriate cases to treat infection and reduce transmission.

Primary sources: NCDC/IDSP vaccine-preventable disease surveillance resources; MoHFW immunization guidance.

14. Influenza and Influenza-like Illness

Cause

Influenza viruses; many other respiratory viruses can cause influenza-like illness.

How it spreads

Respiratory particles/droplets and close contact.

Common symptoms

Sudden fever, cough, sore throat, headache, body aches and fatigue are common.

Warning signs / when to seek urgent care

Breathing difficulty, chest pain, confusion, severe dehydration, bluish lips/face, or worsening illness in a high-risk person needs prompt medical assessment.

Prevention

Vaccination where recommended, respiratory hygiene, hand hygiene, ventilation and avoiding close contact when ill.

General management

Most uncomplicated cases are managed with rest, fluids and symptom control. High-risk or severely ill patients may need early clinician-directed antiviral treatment.

Primary sources: NCDC respiratory illness resources; WHO respiratory infection guidance.

15. Acute Respiratory Infection / Pneumonia

Cause

Can be caused by many viruses or bacteria.

How it spreads

Often through respiratory droplets/particles and close contact, depending on the organism.

Common symptoms

Cough, fever, fast or difficult breathing, chest discomfort and fatigue. Children may show poor feeding, chest indrawing or unusual sleepiness.

Warning signs / when to seek urgent care

Severe breathing difficulty, chest indrawing, blue lips, confusion, inability to drink, low oxygen signs or rapid deterioration require urgent care.

Prevention

Vaccination, hand and respiratory hygiene, reducing smoke exposure and timely treatment of underlying conditions.

General management

Clinical examination is needed to distinguish mild upper respiratory infection from pneumonia or other serious illness. Antibiotics are only appropriate for selected bacterial infections.

Primary sources: MoHFW Management of Acute Respiratory Infection in Children; Medicine (Respiratory) Standard Treatment Guidelines.

Source References

The disease summaries above are paraphrased for RAG use. For clinical decisions, consult the latest official guideline.

MoHFW Standard Treatment Guidelines

<https://clinicalestablishments.mohfw.gov.in/en/standard-treatment-guidelines>

NCDC, Ministry of Health & Family Welfare

<https://ncdc.mohfw.gov.in/>

WHO Chickenpox

<https://www.who.int/news-room/questions-and-answers/item/chickenpox>

WHO Dengue

<https://www.who.int/news-room/fact-sheets/detail/dengue-and-severe-dengue>

WHO Malaria

<https://www.who.int/health-topics/malaria>

WHO Chikungunya

<https://www.who.int/news-room/fact-sheets/detail/chikungunya>

WHO Typhoid

<https://www.who.int/news-room/fact-sheets/detail/typhoid>

WHO Measles

<https://www.who.int/news-room/fact-sheets/detail/measles>

WHO Cholera

<https://www.who.int/news-room/fact-sheets/detail/cholera>

Important: This reference is for general health information. It does not replace examination, diagnosis, laboratory testing, or treatment by a qualified healthcare professional. Emergency symptoms require urgent local medical care.